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CRYSTALLIZATION SYSTEM AND CRYSTALLIZATION METHOD OF LITHIUM HEXAFLUOROPHOSPHATE

Abstract

In the crystallization system of the lithium hexafluorophosphate of the present invention, a first-stage crystallization kettle is connected to a lithium hexafluorophosphate mother liquor supply source through a lithium hexafluorophosphate mother liquor supply line; further connected to the lithium hexafluorophosphate mother liquor supply source through a mother liquor heat exchange line connected with a mother liquor heat exchanger; and further connected to a second-stage crystallization kettle through a first mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor, a second-stage crystallization kettle is connected to a third-stage crystallization kettle through a second mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor, a third-stage crystallization kettle is connected to a fourth-stage crystallization kettle through a crystalline particle delivery line configured to deliver small crystalline particles in the third-stage crystallization kettle; and further connected to a next process device through a first discharging line, and a fourth-stage crystallization kettle is connected to a next process device through a second discharging line. The crystallization system of lithium hexafluorophosphate of the present invention enables different crystallization stages to be carried out separately in different crystallization kettles, so as to realize continuous feeding and continuous discharging in the course of crystallization.

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Background/Summary

[0001] The present application claims the priority of Chinese Patent Application No. 202210915939.5, entitled “CRYSTALLIZATION SYSTEM AND CRYSTALLIZATION METHOD OF LITHIUM HEXAFLUOROPHOSPHATE” and filed on Aug. 1, 2022, which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a production system of lithium hexafluorophosphate, in particular to a crystallization system and crystallization method of lithium hexafluorophosphate.

BACKGROUND

[0003] Lithium hexafluorophosphate, a lithium salt as a key material for lithium-ion batteries, is applied to lithium-ion electrolytes in two main forms: one is crystalline lithium hexafluorophosphate, and the other is a liquid lithium salt formulated together with an organic solvent. Lithium hexafluorophosphates currently sold in foreign markets are all crystalline lithium hexafluorophosphates. The reasons are mainly rooted in benefit to product transportation and preservation with less hidden safety hazards and more compatibility with the process systems of electrolyte manufacturers. With the healthy, sustained, and steady development of the renewable energy industry in China, the demand for crystalline lithium hexafluorophosphates is also increasing. Breakthrough and improvement of the production technology of crystalline lithium hexafluorophosphates and improvement of the product quality of crystalline lithium hexafluorophosphates are the focus of domestic research at present.

[0004] At present, the preparation method for lithium hexafluorophosphate mainly includes: a gas-solid direct reaction method, a solvent method, and an ion exchange method, among which the most extensively researched, technically mature, and widely used technique in industrialization is the hydrogen fluoride solvent method. The currently available hydrogen fluoride solvent method is to dissolve a lithium salt in anhydrous hydrofluoric acid to form a LiFHF solution, into which phosphorus pentafluoride (PF₅) gas is introduced for reaction to yield lithium hexafluorophosphate. That is, the technology of preparing lithium hexafluorophosphate by the hydrogen fluoride solvent method is currently a more mature technology and is also the easiest production method to realize industrialization.

[0005] Specifically, the hydrogen fluoride solvent method is a method of using hydrogen fluoride as a reaction medium, dissolving raw material lithium halide in the hydrogen fluoride, then gasifying high-purity phosphorus pentafluoride, and next introducing gas into the solvent for reaction to yield a lithium hexafluorophosphate crystal. After completion of the reaction, the lithium hexafluorophosphate crystal is subjected to crystal separation, drying and the like to finally

obtain a lithium hexafluorophosphate product.

[0006] In spite of highly concerned, lithium hexafluorophosphates are still unable to be produced on a large scale mainly because of the great difficulties in production technology.

[0007] As the commonly used crystallization method for lithium hexafluorophosphate, there are mainly two methods: static crystallization and agitated crystallization. The two methods have their own advantages and disadvantages. The static crystallization includes the following advantages: (1) hydrogen fluoride is relatively in a static state, so it is more stable and safer; and (2) the crystallization speed is slow and products and impurities can be separated and precipitated in the crystalline solution due to their difference in solubility, which is conducive to the purity and quality of the finished products. The static crystallization includes the following disadvantages: (1) there is a need for a sufficiently low temperature and stricter requirements for freezing, and it takes long time to crystallize, resulting in large energy consumption, which is not conducive to cost control; and (2) the particles resulting from crystallization are large and need to be further crushed by a physical method, which leads to irregular and non-uniform particles of the product and poorer fluidity. Besides, the agitated crystallization includes the following advantages: (1) the agitated crystallization allows for uniform particles and relatively stable product quality; and (2) the crystallization time is short, the requirements for temperature are lower than those for the static crystallization, the energy consumption is less, and its production cycle is shorter than that of the static crystallization, which can effectively improve the production efficiency. Nevertheless, the agitated crystallization includes the following disadvantages: (1) agitated crystallization of hydrofluoric acid causes volatility to increase, which leads to increased loss of hydrogen fluoride raw material; (2) the crystallization speed of the agitated crystallization is faster, which may lead to crystallization of impurity components in the mother liquor together, resulting in more impurity components in the product; and (3) mechanical seal of the agitation is prone to corrosion and leakage.

[0008] In recent years, the agitated crystallization is widely used. However, the current agitated crystallization is always batch crystallization, that is, the synthesized mother liquor of lithium hexafluorophosphates is subjected to crystallization, grain growth, and final growth to the desired size of grains in the same crystallization kettle. The mother liquor of lithium hexafluorophosphate undergoes several different stages from crystallization to growth to the desired sized of grains, and each of the stages is required to control the crystallization kettle to adopt different process parameters (such as temperature). In this way, not only is the control more complicated, but the course of crystallization fails to realize continuous feeding (feeding of mother liquors) and continuous discharging (delivering the lithium hexafluorophosphate grains finally growing to the desired size into the next process device). Consequently, the course of crystallization including the feeding process and discharging process still has the problems such as long cycle and low turnover efficiency of devices such as a crystallization kettle.

[0009] In view of the above, there is now an urgent need in this technical field for a crystallization system and crystallization method of lithium hexafluorophosphate that can improve the turnover efficiency of devices such as a crystallization kettle and further improve the production efficiency of the agitated crystallization.

SUMMARY

[0010] A crystallization system and crystallization method of lithium hexafluorophosphate provided in the present invention are designed, on the basis of agitated crystallization, to carry out different crystallization stages separately in different crystallization kettles according to the crystallization characteristics of lithium hexafluorophosphate, and connect the crystallization kettles for carrying out different crystallization stages in series, such that continuous feeding and continuous discharging are realized in the course of crystallization, thereby fulfilling the purpose of further improving the production efficiency of lithium hexafluorophosphate.

[0011] The present invention provides a crystallization system of lithium hexafluorophosphate,

comprising a first-stage crystallization kettle, a second-stage crystallization kettle, a third-stage crystallization kettle, and one or more fourth-stage crystallization kettles: the first-stage crystallization kettle, which is connected to a lithium hexafluorophosphate mother liquor supply source through a lithium hexafluorophosphate mother liquor supply line: further connected to the lithium hexafluorophosphate mother liquor supply source through a mother liquor heat exchange line connected with a mother liquor heat exchanger; and further connected to the second-stage crystallization kettle through a first mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor having a first prescribed grain size, the second-stage crystallization kettle, which is connected to the third-stage crystallization kettle through a second mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor having a second prescribed grain size, the third-stage crystallization kettle, which is connected to the fourth-stage crystallization kettle through a crystalline particle delivery line configured to deliver small crystalline particles in the third-stage crystallization kettle; and further connected to a next process device through a first discharging line: the fourth-stage crystallization kettle, which is connected to a next process device through a second discharging line: when the lithium hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source to the first-stage crystallization kettle crystallizes and grows to the first prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a first prescribed temperature range by the mother liquor heat exchanger, the lithium hexafluorophosphate mother liquor being delivered to the second-stage crystallization kettle through the first mother liquor delivery line: when the lithium hexafluorophosphate mother liquor delivered to the second-stage crystallization kettle continues to crystallize and grow to the second prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a second prescribed temperature range, the lithium hexafluorophosphate mother liquor being delivered to the third-stage crystallization kettle through the second mother liquor delivery line: a part of the lithium hexafluorophosphate mother liquor delivered to the third-stage crystallization kettle growing to crystal particles having a third prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a third prescribed temperature range, and the crystal particles having a third prescribed grain size being delivered to a next process device through the first discharging line, while the other part grow to crystal particles having a grain size less than the third prescribed grain size, and the crystal particles having a grain size less than the third prescribed grain size are delivered to the fourth-stage crystallization kettle through the crystalline particle delivery line; and lithium hexafluorophosphate crystal particles delivered to the fourth-stage crystallization kettle growing to crystal particles having a fourth prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate crystals is controlled within a fourth prescribed temperature range, and the crystal particles having a fourth prescribed grain size being delivered to a next process device through the second discharging line.

[0012] Preferably, the first prescribed temperature range is from -5°C. to 0°C. , and the first prescribed grain size is from $10\text{ }\mu\text{m}$ to $50\text{ }\mu\text{m}$: the second prescribed temperature range is from -15°C. to -10°C. , and the second prescribed grain size is from $100\text{ }\mu\text{m}$ to $150\text{ }\mu\text{m}$; the third prescribed temperature range is from -30°C. to -25°C. , and the third prescribed grain size is from $180\text{ }\mu\text{m}$ to $200\text{ }\mu\text{m}$; and the fourth prescribed temperature range is from -32°C. to -25°C. , and the fourth prescribed grain size is from $180\text{ }\mu\text{m}$ to $200\text{ }\mu\text{m}$.

[0013] Preferably, the first-stage crystallization kettle, the second-stage crystallization kettle, the third-stage crystallization kettle, and at least one crystallization kettle of the fourth-stage crystallization kettles are provided with an agitator, and a temperature drop rate that the temperature in the crystallization kettle at each of stages drops to the prescribed temperature range is controlled.

[0014] Additionally preferably, assuming that a mother liquor feeding amount of the lithium

hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source to the first-stage crystallization kettle is Q1, a first discharging amount of the lithium hexafluorophosphate crystal particles delivered to the next process device through the first discharging line is Q2, and a second discharging amount of the lithium hexafluorophosphate crystal particles delivered to the next process device through the second discharging line is Q3, the mother liquor feeding amount Q1 is a sum of the first discharging amount Q2 and the second discharging amount, and the first discharging amount Q2 is twice the second discharging amount Q3.

[0015] Further preferably, the lithium hexafluorophosphate mother liquor supply line is provided with a flow meter and a regulating valve; the mother liquor heat exchange line is provided with an axial-flow pump; and the first mother liquor delivery line, the second mother liquor delivery line, and the crystalline particle delivery line are provided with a regulating valve, respectively.

[0016] Preferably, the lithium hexafluorophosphate mother liquor supply line is connected to the mother liquor heat exchange line.

[0017] The present invention further provides a crystallization method of lithium hexafluorophosphate, which is a crystallization method for crystallizing a mother liquor of the lithium hexafluorophosphate using the crystallization system of lithium hexafluorophosphate described above, comprising: a first-stage crystallization process carried out in the first-stage crystallization kettle, a second-stage crystallization process carried out in the second-stage crystallization kettle, a third-stage crystallization process carried out in the third-stage crystallization kettle, and a fourth-stage crystallization process carried out in the fourth-stage crystallization kettle, the crystallization processes from the first-stage crystallization process through the second-stage crystallization process to the third-stage crystallization process and the fourth-stage crystallization process being carried out continuously.

[0018] The crystallization system and crystallization method of lithium hexafluorophosphate according to the present invention may allow different crystallization stages to be carried out separately in different crystallization kettles and allow the crystallization kettles for carrying out different crystallization stages to be connected in series, such that continuous feeding and continuous discharging are realized in the course of crystallization, thereby fulfilling the purpose of further improving the production efficiency of lithium hexafluorophosphate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 shows a schematic diagram of a crystallization system of lithium hexafluorophosphate according to the present invention.

[0020] FIG. 2 shows a process flowchart of a crystallization system of lithium hexafluorophosphate according to the present invention.

[0021] FIG. 3 shows a process flowchart of another example of a crystallization system of lithium hexafluorophosphate according to the present invention.

[0022] In the drawings, **10**-crystallization system of lithium hexafluorophosphate: **11**-first-stage crystallization kettle: **12**-second-stage crystallization kettle: **13**-third-stage crystallization kettle: **14**-fourth-stage crystallization kettle: **20**-lithium hexafluorophosphate mother liquor supply source: **30**-next process device.

DETAILED DESCRIPTION

[0023] The technical solutions of the present invention and effects thereof will be described in detail below with reference to specific embodiments. The following embodiments are used only for explaining the content of the present invention, but the invention is not just limited to the following embodiments or examples. All simple changes made to the present invention based on the concept of the present invention fall within the scope of protection for the present invention.

[0024] A crystallization system and crystallization method of lithium hexafluorophosphate provided in the present invention are designed, on the basis of agitated crystallization, to carry out different crystallization stages separately in different crystallization kettles according to the crystallization characteristics of lithium hexafluorophosphate, and connect the crystallization kettles for carrying out different crystallization stages in series, such that continuous feeding and continuous discharging are realized in the course of crystallization, thereby fulfilling the purpose of further improving the production efficiency of lithium hexafluorophosphate.

[0025] Specifically, a crystallization system **10** of lithium hexafluorophosphate provided in the present invention (as shown in FIG. **1**) comprises a first-stage crystallization kettle **11**, a second-stage crystallization kettle **12**, a third-stage crystallization kettle **13**, and a fourth-stage crystallization kettle **14**.

[0026] The first-stage crystallization kettle **11** is connected to a lithium hexafluorophosphate mother liquor supply source **20** through a lithium hexafluorophosphate mother liquor supply line **11a**, and a flow meter **11b** and a regulating valve **11c** are provided on the lithium hexafluorophosphate mother liquor supply line **11a**.

[0027] Furthermore, the first-stage crystallization kettle **11** is further connected to the lithium hexafluorophosphate mother liquor supply source **20** through a mother liquor heat exchange line **11e** connected with a mother liquor heat exchanger **11d**, and the mother liquor heat exchange line **11e** is provided with an axial-flow pump **11f**. Concretely, one end of the mother liquor heat exchange line **11e** is communicated with the first-stage crystallization kettle **11**, and the other end of the mother liquor heat exchange line **11e** is communicated with the lithium hexafluorophosphate mother liquor supply line **11a**.

[0028] In addition, the first-stage crystallization kettle **11** is further connected to the second-stage crystallization kettle **12** through a first mother liquor delivery line **11g** configured to deliver lithium hexafluorophosphate mother liquor having a first prescribed grain size, and the first mother liquor delivery line **11g** is provided with a regulating valve **11h**.

[0029] The second-stage crystallization kettle **12** is connected to the third-stage crystallization kettle **13** through a second mother liquor delivery line **12a** configured to deliver lithium hexafluorophosphate mother liquor having a second prescribed grain size, and the second mother liquor delivery line **12a** is provided with a regulating valve **12b**.

[0030] The third-stage crystallization kettle **13** is connected to the fourth-stage crystallization kettle **14** through a crystalline particle delivery line **13a** configured to deliver small crystalline particles in the third-stage crystallization kettle **13**, and the crystalline particle delivery line **13a** is provided with a regulating valve **13b**. Besides, the third-stage crystallization kettle **13** is further connected to a next process device **30** through a first discharging line **13c**, and the first discharging line **13c** is provided with a metering pump **13d**.

[0031] The fourth-stage crystallization kettle **14** is connected with a next process device **30** (such as a crystallization drying device) through a second discharging line **14a**, and the second discharging line **14a** is provided with a metering pump **14b**.

[0032] In addition, an agitator **40** is provided in the first-stage crystallization kettle **11**, the second-stage crystallization kettle **12**, the third-stage crystallization kettle **13**, and the fourth-stage crystallization kettle **14**, respectively.

[0033] Hereinafter, the process flow of the crystallization system **10** of lithium hexafluorophosphate of the present invention is described with reference to FIG. **1** and FIG. **2**.

[0034] As shown in FIG. **1** and FIG. **2**, when the lithium hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source **20** to the first-stage crystallization kettle **11** crystallizes and grows to the first prescribed grain size (10 to 50 μm in this embodiment) under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within the first prescribed temperature range (-5°C . to 0°C . in this embodiment) by the mother liquor heat exchanger **11d**, the lithium hexafluorophosphate mother

liquor is delivered to the second-stage crystallization kettle **12** through the first mother liquor delivery line **11g**. When the lithium hexafluorophosphate mother liquor delivered to the second-stage crystallization kettle **12** continues to crystallize and grows to the second prescribed grain size (100 to 150 μm in this embodiment) under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within the second prescribed temperature range (-15°C . to -10°C . in this embodiment), the lithium hexafluorophosphate mother liquor is delivered to the third-stage crystallization kettle **13** through the second mother liquor delivery line **12a**. A part of the lithium hexafluorophosphate mother liquor delivered to the third-stage crystallization kettle **13** grows to crystal particles having the third prescribed grain size (180 to 200 μm in this embodiment) under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within the third prescribed temperature range (-30°C . to -25°C . in this embodiment), and the crystal particles having the third prescribed grain size are delivered to a next process device **30** through the first discharging line **13c**, while the other part grows to crystal particles having a grain size less than the third prescribed grain size and the crystal particles having a grain size less than the third prescribed grain size are delivered to the fourth-stage crystallization kettle **14** through the crystalline particle delivery line **13a**. The lithium hexafluorophosphate crystal particles delivered to the fourth-stage crystallization kettle **14** grows to crystal particles having the fourth prescribed grain size (180 to 200 μm in this embodiment) under the condition that the temperature of the lithium hexafluorophosphate crystal particles is controlled within the fourth prescribed temperature range (-32°C . to -25°C . in this embodiment), and the crystal particles having the fourth prescribed grain size are delivered to a next process device **30** through the second discharging line **14a**.

[0035] Additionally, during the crystallization in the first-stage crystallization kettle **11**, the temperature in the first-stage crystallization kettle **11** is controlled to slowly drop to the first prescribed temperature range (-5°C . to 0°C . in this embodiment). In this way, more uniform small crystals are easily precipitated in the first-stage crystallization kettle **11**, and the amount of crystals precipitated is controllable without excessive small crystals. Moreover, the small crystals are dispersed by agitating with an agitator **40** to reduce the probability for the small crystals to encounter each other and lessen aggregation and settlement resulting from accumulation of small crystals, thereby ensuring that the small crystals supplied from the first-stage crystallization kettle **11** to the second-stage crystallization kettle **12** are uniform, which contributes to improvement of the crystallization efficiency and acquisition of uniform crystals. During the crystallization in the second-stage crystallization kettle **12**, the temperature in the second-stage crystallization kettle **12** is controlled to rapidly drop to the second prescribed temperature range (-15°C . to -10°C . in this embodiment). In this way, the small crystals from the first-stage crystallization kettle **11** grow rapidly and uniformly in the second-stage crystallization kettle **12** along its surface. Moreover, the growing crystals are dispersed by agitating with the agitator **40** to reduce the probability for the grown crystals to encounter each other and lessen aggregation and settlement resulting from accumulation of crystals, thereby ensuring that the crystals supplied from the second-stage crystallization kettle **12** to the third-stage crystallization kettle **13** are uniform, which contributes to improvement of the crystallization efficiency and acquisition of uniform crystals. In the third-stage crystallization kettle **13**, grains having a large grain size are settled in the downstream, and grains having a small grain size are settled in the midstream and upstream and will enter the fourth-stage crystallization kettle **14** through the line. The temperature in the fourth-stage crystallization kettle **14** may be set to be lower (about 2°C .) than that in the third-stage crystallization kettle **13**. In this way, a part of crystals will be further precipitated in the fourth-stage reaction kettle **14** and grow on the surfaces of the grains having a small grain size. Finally, the crystals in the third-stage crystallization kettle **13** and the fourth-stage crystallization kettle **14** are precipitated together with homogeneous grain sizes.

[0036] In addition, assuming that a mother liquor feeding amount of the lithium

hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source to the first-stage crystallization kettle is **Q1**, a first discharging amount of the lithium hexafluorophosphate crystal particles delivered to the next process device through the first discharging line is **Q2**, and a second discharging amount of the lithium hexafluorophosphate crystal particles delivered to the next process device through the second discharging line is **Q3**, the mother liquor feeding amount **Q1** is a sum of the first discharging amount **Q2** and the second discharging amount, and the first discharging amount **Q2** is twice the second discharging amount **Q3**.

[0037] The crystallization method of lithium hexafluorophosphate provided in the present invention, which is a crystallization method for crystallizing a mother liquor of the lithium hexafluorophosphate using the crystallization system **10** of lithium hexafluorophosphate described above, comprises: a first-stage crystallization process carried out in the first-stage crystallization kettle **11**, a second-stage crystallization process carried out in the second-stage crystallization kettle **12**, a third-stage crystallization process carried out in the third-stage crystallization kettle **13**, and a fourth-stage crystallization process carried out in the fourth-stage crystallization kettle **14**, the crystallization processes from the first-stage crystallization process through the second-stage crystallization process to the third-stage crystallization process and the fourth-stage crystallization process being carried out continuously.

[0038] The crystallization system and crystallization method of lithium hexafluorophosphate according to the present invention may allow different crystallization stages to be carried out separately in different crystallization kettles and allow the crystallization kettles for carrying out different crystallization stages to be connected in series, such that continuous feeding and continuous discharging are realized in the course of crystallization, thereby fulfilling the purpose of further improving the production efficiency of lithium hexafluorophosphate. Meanwhile, the crystallization in the crystallization kettles at all stages are precisely controlled to reduce aggregation and settlement during the crystallization, so as to ensure that crystals in the crystallization kettles at all stages are uniform, thereby improving the crystallization efficiency and obtaining uniform crystals.

[0039] As other embodiments of the present invention, the structure and process flow as shown in FIG. **3** may be used, for example. In FIG. **3**, arrows indicate the flow directions of materials (mother liquors or crystalline particles). This embodiment differs from the above embodiments in that another fourth-stage crystallization kettle **15** is further provided in addition to the fourth-stage crystallization kettle **14**. That is, in this case, a part of the lithium hexafluorophosphate mother liquor delivered to the third-stage crystallization kettle **13** grows to crystal particles having the third prescribed grain size (180 to 200 μm in this embodiment) under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within the third prescribed temperature range (-30°C. to -25°C. in this embodiment), and the crystal particles having the third prescribed grain size are delivered to a next process device **30** through the first discharging line **13c**, while the other part grows to crystal particles having a grain size less than the third prescribed grain size, and the crystal particles having a grain size less than the third prescribed grain size are further delivered to the fourth-stage crystallization kettle **15** through another crystalline particle delivery line (not shown) in addition to being delivered to the fourth-stage crystallization kettle **14** through the crystalline particle delivery line **13a**. Besides, three or more fourth-stage crystallization kettles may alternatively be provided.

[0040] Further, it is to be noted that each of the specific technical features described in the above specific embodiments may be combined in any suitable manner if without contradiction.

Claims

1. A crystallization system of lithium hexafluorophosphate, comprising a first-stage crystallization kettle, a second-stage crystallization kettle, a third-stage crystallization kettle, and one or more

fourth-stage crystallization kettles; the first-stage crystallization kettle, which is connected to a lithium hexafluorophosphate mother liquor supply source through a lithium hexafluorophosphate mother liquor supply line; further connected to the lithium hexafluorophosphate mother liquor supply source through a mother liquor heat exchange line connected with a mother liquor heat exchanger; and further connected to the second-stage crystallization kettle through a first mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor having a first prescribed grain size, the second-stage crystallization kettle, which is connected to the third-stage crystallization kettle through a second mother liquor delivery line configured to deliver lithium hexafluorophosphate mother liquor having a second prescribed grain size, the third-stage crystallization kettle, which is connected to the fourth-stage crystallization kettle through a crystalline particle delivery line configured to deliver small crystalline particles in the third-stage crystallization kettle; and further connected to a next process device through a first discharging line; the fourth-stage crystallization kettle, which is connected to a next process device through a second discharging line; when the lithium hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source to the first-stage crystallization kettle crystallizes and grows to the first prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a first prescribed temperature range by the mother liquor heat exchanger, the lithium hexafluorophosphate mother liquor being delivered to the second-stage crystallization kettle through the first mother liquor delivery line; when the lithium hexafluorophosphate mother liquor delivered to the second-stage crystallization kettle continues to crystallize and grow to the second prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a second prescribed temperature range, the lithium hexafluorophosphate mother liquor being delivered to the third-stage crystallization kettle through the second mother liquor delivery line; a part of the lithium hexafluorophosphate mother liquor delivered to the third-stage crystallization kettle growing to crystal particles having a third prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate mother liquor is controlled within a third prescribed temperature range, and the crystal particles having a third prescribed grain size being delivered to a next process device through the first discharging line, while the other part grows to crystal particles having a grain size less than the third prescribed grain size, and the crystal particles having a grain size less than the third prescribed grain size are delivered to the fourth-stage crystallization kettle through the crystalline particle delivery line; and the lithium hexafluorophosphate crystal particles delivered to the fourth-stage crystallization kettle grow to crystal particles having a fourth prescribed grain size under the condition that the temperature of the lithium hexafluorophosphate crystal particles is controlled within a fourth prescribed temperature range, and the crystal particles having a fourth prescribed grain size being delivered to a next process device through the second discharging line.

2. The crystallization system of lithium hexafluorophosphate according to claim 1, wherein the first prescribed temperature range is from -5°C. to 0°C. , and the first prescribed grain size is from $10\text{ }\mu\text{m}$ to $50\text{ }\mu\text{m}$; the second prescribed temperature range is from -15°C. to -10°C. , and the second prescribed grain size is from $100\text{ }\mu\text{m}$ to $150\text{ }\mu\text{m}$; the third prescribed temperature range is from -30°C. to -25°C. , and the third prescribed grain size is from $180\text{ }\mu\text{m}$ to $200\text{ }\mu\text{m}$; and the fourth prescribed temperature range is from -32°C. to -25°C. , and the fourth prescribed grain size is from $180\text{ }\mu\text{m}$ to $200\text{ }\mu\text{m}$.

3. The crystallization system of lithium hexafluorophosphate according to claim 1, wherein the first-stage crystallization kettle, the second-stage crystallization kettle, the third-stage crystallization kettle, and at least one crystallization kettle of the fourth-stage crystallization kettles are provided with an agitator, and a temperature drop rate that the temperature in the crystallization kettle at each of stages drops to the prescribed temperature range is controlled.

4. The crystallization system of lithium hexafluorophosphate according to claim 1, wherein

assuming that a mother liquor feeding amount of the lithium hexafluorophosphate mother liquor supplied from the lithium hexafluorophosphate mother liquor supply source to the first-stage crystallization kettle is Q_1 , a first discharging amount of the lithium hexafluorophosphate crystals delivered to the next process device through the first discharging line is Q_2 , and a second discharging amount of the lithium hexafluorophosphate crystals delivered to the next process device through the second discharging line is Q_3 , the mother liquor feeding amount Q_1 is a sum of the first discharging amount Q_2 and the second discharging amount Q_3 , and the first discharging amount Q_2 is twice the second discharging amount Q_3 .

5. The crystallization system of lithium hexafluorophosphate according to claim 1, wherein the lithium hexafluorophosphate mother liquor supply line is provided with a flow meter and a regulating valve; the mother liquor heat exchange line is provided with a axial-flow pump; and the first mother liquor delivery line, the second mother liquor delivery line, and the crystalline particle delivery line are provided with a regulating valve, respectively.

6. The crystallization system of lithium hexafluorophosphate according to claim 5, wherein the lithium hexafluorophosphate mother liquor supply line is connected to the mother liquor heat exchange line.

7. A crystallization method of lithium hexafluorophosphate, which is a crystallization method for crystallizing a mother liquor of the lithium hexafluorophosphate using a crystallization system of lithium hexafluorophosphate according to claim 1, comprising: a first-stage crystallization process carried out in the first-stage crystallization kettle, a second-stage crystallization process carried out in the second-stage crystallization kettle, a third-stage crystallization process carried out in the third-stage crystallization kettle, and a fourth-stage crystallization process carried out in the fourth-stage crystallization kettle, the crystallization processes from the first-stage crystallization process through the second-stage crystallization process to the third-stage crystallization process and the fourth-stage crystallization process being carried out continuously.
